θ
$X_1, X_2, \ldots, X_n$ ” ”
θ
data reduction
$\overline{X}$
S^2
θ
sufficiency
$T(X_1, \ldots, X_n)$
θ
T
θ
$\Rightarrow$
$\Rightarrow$
$\Rightarrow$
$\Rightarrow$
$Y =$
$u(X_1, X_2, \ldots, X_n)$
n
θ
Y
$E(Y) =$
θ
θ
Y'
$var(Y) \leq$
$var(Y')$
Y
θ
minimum variance unbiased estimator, MVUE
θ
θ
$X_1, X_2, \ldots, X_9$
$N(\theta, \sigma^2)$
$-\infty <$
$\theta <$
∞
$\overline{X} =$
$(X_1 +$
$X_2 +$
$\cdots +$
$X_9)/9$
$N(\theta, \sigma^2/9)$
X_1
$N(\theta, \sigma^2)$
$\overline{X}$
$\sigma^2/9$
X_1
σ^2
$\overline{X}$
$\mathcal{L}[\theta, \delta(y)]$
$\delta(y)$
θ
$\mathcal{L}[\theta, \delta(y)] =$
$[\theta -$
$\delta(y)]^2$
$R(\theta, \delta) =$
$E[\mathcal{L}(\theta, \delta(Y))]$
$R(\theta, \delta)$
θ
$X_1, \ldots, X_{25}$
$N(\theta, 1)$
$\mathcal{L}[\theta, \delta(y)] =$
$[\theta -$
$\delta(y)]^2$
$\delta_1(y) =$
$y =$
$\overline{x}$
$R(\theta, \delta_1) =$
$1/25$
$\delta_2(y) =$
0
$R(\theta, \delta_2) =$
θ^2
$\theta =$
0
δ_2
$R(0, \delta_2) =$
0
$\theta =$
2
$R(2, \delta_2) =$
$4 >$